Jordan Dowdy

PhD Candidate | University of Louisville
Member, **Telos Lab**

CURRICULUM VITAE

ROBOT LEARNING
REINFORCEMENT LEARNING
EMBODIED SYSTEMS

Google Scholar | GitHub | LinkedIn | X / Twitter

RESEARCH PROFILE

PhD candidate researching reinforcement learning for embodied robots, with work spanning legged locomotion, sim-to-real transfer, tactile sensing, and physical human-robot interaction. Research emphasizes methods that remain useful under changing tasks, operating conditions, and deployment constraints.

RESEARCH AREAS

Reinforcement learning | Robot learning | Legged locomotion | Sim-to-real transfer | Tactile sensing | Physical human-robot interaction

RESEARCH HIGHLIGHTS

01 Torque-driven and sim-to-real reinforcement-learning methods for quadruped locomotion across rough terrain, stairs, and inclines.

02 Reinforcement-learning locomotion paired with virtual-reality upper-body control for miniature humanoid tele-loco-manipulation.

03 Tactile calibration, neuroadaptive control, and robotic-skin sensing for physical human-robot collaboration.

EDUCATION AND AFFILIATION

CURRENT **University of Louisville**
PhD candidate conducting robotics research in reinforcement learning; member of Telos Lab.

2024 **University of Louisville**
M.S. thesis in Electrical Engineering: *Adaptive Robot Collaboration Using Robotic Skin and Motion Similarity*.

SELECTED PUBLICATIONS

Real-Time Neuroadaptive Control with Tactile Calibration for Physical Human-Robot Interaction
Ashutosh Prakash, Mohamed A. Hanafy, **Jordan Dowdy**, Dan O. Popa
2026 | Preprint (not peer-reviewed) | Preprints.org, Version 1

Towards Torque-Driven Reinforcement Learning for Quadruped Locomotion
Jordan Dowdy, Jean Chagas Vaz
2026 | Conference paper | IEEE/SICE International Symposium on System Integration (SII), pp. 1259-1264

Towards Miniature Humanoid Tele-Loco-Manipulation Using Virtual Reality and Reinforcement Learning
Nicolas Kosanovic, **Jordan Dowdy**, Jean Chagas Vaz
2025 | Conference paper | IEEE-RAS International Conference on Humanoid Robots, pp. 1233-1240

Isaac Sim-to-Real: Reinforcement Learning based Locomotion for Quadrupeds
Jordan Dowdy, Jean Chagas Vaz
2025 | Conference paper | IEEE International Conference on Automation Science and Engineering (CASE), pp. 2194-2199

Adaptive Robot Collaboration Using Robotic Skin and Motion Similarity
Jordan Dowdy
2024 | M.S. thesis | Electrical Engineering, University of Louisville